

PAPER • OPEN ACCESS

Sol-gel fabrication of one-dimensional photonic crystals with predicted transmission spectra

To cite this article: V A Il'ykh and L B Matyushkin 2016 *J. Phys.: Conf. Ser.* **741** 012008

View the [article online](#) for updates and enhancements.

You may also like

- [Experimental sol-gel fabrication and theoretical simulation of one-dimensional photonic crystals with a defect state](#)
V A Romanova, P A Somov and L B Matyushkin

- [Optical dispersion in spun nanocrystalline titania thin films](#)
R Capan, N B Chaure, A K Hassan et al.

- [Optical transmission spectra in quasiperiodic multilayered photonic structure](#)
F F de Medeiros, E L Albuquerque and M S Vasconcelos

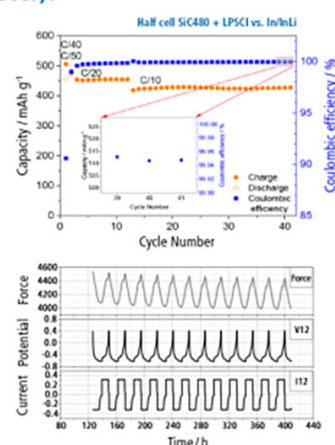
The New PAT-Cell-Solid!

Cycle Solid-State Batteries Under Controlled Pressure of up to 300 MPa (6 mm Diameter)!



- ✓ **Adjust and measure a force of up to 9000 N on the cell stack!**
Force adjustment possible throughout the entire experiment
- ✓ **Built-in force, and temperature sensors!**
With optional gas pressure sensor and gas in- and outlet
- ✓ **PAT-Solid-Core for easy assembly and reproducible results!**
Press and cycle solid-state batteries with 6 or 10 mm electrode diameter
- ✓ **Cableless and highly sealed battery test cell!**
For precise long-term measurements of solid-state cell chemistries

EL-CELL®
electrochemical test equipment



Learn more on our product website:



Scan me!

Download the data sheet (PDF):



Scan me!

Or contact us directly:

+49 40 79012-734

sales@el-cell.com

www.el-cell.com

Sol-gel fabrication of one-dimensional photonic crystals with predicted transmission spectra

V A Ilinykh¹, L B Matyushkin¹

¹Department of Micro- and Nanoelectronics, St. Petersburg State Electrotechnical University "LETI", St. Petersburg, 197376, Russia

Abstract. One-dimensional multilayer structures of periodically alternating low refractive index (silica) and high refractive index (titania) materials have been deposited by sol-gel spin-coating. Experimental spectra of the structures are in agreement with spectra calculated by transfer matrix technique. As an example, theoretical and experimental spectra with a stop band corresponding 600 nm-reflection are shown.

1. Introduction

Photonic bandgap structures (PBGs), also known as photonic crystals (PCs), are structures whose refractive index is periodic on a length scale of the order of optical wavelengths, which prevents light propagating through the structure due to Bragg reflection. Depending on the dimensionality of this periodicity, PBGs can be one, two or three dimensions (1D, 2D or 3D). In the case of sol-gel derived materials, the main emphasis has been on 1D- and 3D-structures [1]. Distributed Bragg reflectors (DBR), also called Bragg mirrors, are examples of 1D-PBGs. Their overall reflectivity can be very high (higher than that of metallic mirrors) and it's due to Bragg reflection. Sol-gel DBRs can be made, for example, by dip-coating [2] or spin-coating [3]. The refractive index contrast and the number of layers are the most important parameters. Silica (SiO₂) and titanic (TiO₂) are chosen because of the high optical contrast of their refractive indices (~1.46 and ~2.4, respectively). The higher the refractive indices contrast, the wider the 'stop band'. And the higher the number of layers, the higher reflectivity within the stop band. The application of the sol-gel process for the preparation of photonic crystals dealt with a number of issues described in this paper.

2. Math apparatus

Different 1D-PCs composed of SiO₂/TiO₂ alternating layers were analysed. To calculate multilayer structure transmission spectra the transfer matrix method was used. Matrix A describes light propagation between the layers and matrix B describes light transmission through current material layer:

$$A = \frac{1}{2n_2} \begin{pmatrix} n_1 + n_2 & n_2 - n_1 \\ n_2 - n_1 & n_1 + n_2 \end{pmatrix}, \quad B = \begin{pmatrix} e^{2\pi i \frac{nL}{\lambda}} & 0 \\ 0 & e^{-2\pi i \frac{nL}{\lambda}} \end{pmatrix},$$

where n_1 and n_2 – refractive indices of alternating materials, n , L – refractive index and thickness of current layer, λ – wavelength of light. Elements of alternate matrix multiplication M ($M = A_1 B_1 A_2 B_2 \dots A_n B_n$) allows to describe the optical properties of the multilayer structures.



3. Experimental section

During the preliminary experiments, the film deposition rate, the annealing temperature and the concentration of the starting sols were optimised. Homogeneous films deposited by spin-coating 100 μL of sol are obtained at speeds of about 3500 rpm. PCs were built by deposition of TiO_2 and SiO_2 on soda-lime glass substrates. The first layer was deposited of TiO_2 sol, because SiO_2 and soda-lime glass refractive indices are almost equal. It has been found that the optimum temperature for annealing of the samples is 550°C , the duration of annealing layers is 60 min for titania, and 30 min for silica. The process is repeated until a total number of layers have been deposited.

4. Results and discussion

To investigate producing of photonic crystals by sol-gel technology it have been researched 1D-PC structures with different numbers and thicknesses of layers. Fig. 1a shows the transmission spectra of the example, the numbers indicate the number of layers. Theoretical transmission spectra (Fig.1b) differs slightly from the experimental spectra due some porosity of layers, but shape and stop-band position are the same.

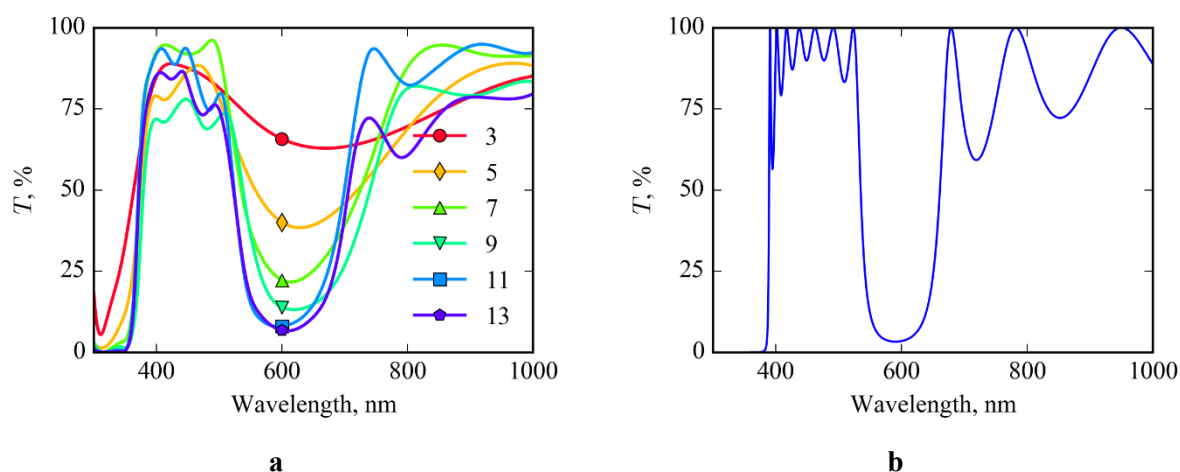


Figure 1(a, b). (a) Transmission spectra of photonic crystals with different number of alternating layers and different thicknesses of layers: SiO_2 – 70 nm, TiO_2 – 100 nm; (b) corresponding theoretical transmission spectrum for 13-layers structure

5. Conclusion

As a result of experiments, one-dimensional photonic crystals was produced under various process conditions. The possibility of tuning of layer thickness by diluting initial sols is shown. Based on described technique, the photonic crystal structures were made with a minimum of transmittance about 7% at the photonic band gaps (450 and 600 nm). Photonic band gap position was varied by changing the viscosity of the solution and the corresponding changes in the thickness of material layers.

Acknowledgments

This study was supported by the Ministry of Education and Science of the Russian Federation within the framework of the project part of the state assignment for the St. Petersburg Electrotechnical University “LETI” (project 16.2112.2014/K, “Preparation and study of porous systems, functionalized nanomaterials, applications in photonics, sensor technology and medicine”).

References

- [1] Almeida R M, Sabine P 2003 *Curr. Opin. Solid State Mater. Sci.* **7(2)** 151
- [2] Bellessa J, Rabaste S, Plenet J C, Dumas J, Mugnier J, Marty O 2001 *Appl. Phys. Lett.* **79** 2142
- [3] Min, Xin, Xavier Orignac, and Rui M. Almeida 1995 *J. Am. Ceram. Soc.* **78(8)** 2254